Pramo Jewels

3D Design Guidelines

Document ID: PJ-3D-001
Version: 1.0
Status: Draft

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define standards for all 3D assets and interactions
Audience	3D Artists, UI Designers, Frontend Developers, AI Coding Agents
Version	1.0

1. 3D Design Philosophy

3D elements should reinforce the premium identity of Pramo Jewels by showcasing jewellery with realistic materials, refined motion and subtle interactions. 3D should enhance usability rather than distract from it.

2. Design Principles

• Product-first presentation.
• Photorealistic rendering.
• Minimal visual clutter.
• Smooth interaction.
• Performance-conscious implementation.

3. Materials

• Physically Based Rendering (PBR).
• Accurate gold, silver and gemstone materials.
• Realistic reflections and roughness values.
• High-quality normal and ambient occlusion maps where required.

4. Lighting

• HDRI studio lighting.
• Soft key, fill and rim lights.
• Controlled reflections.
• Neutral white balance to preserve metal colors.

5. Camera Guidelines

• Default three-quarter view.
• Smooth orbit controls.
• Controlled zoom limits.
• Focus on product details without distortion.

6. User Interaction

• 360° rotation.
• Zoom and pan.
• Variant switching with smooth transitions.
• Mouse/touch-driven lighting response where appropriate.

7. Asset Standards

Preferred format: GLB/GLTF.
Optimized mesh topology.
Compressed textures.
Consistent naming by SKU and collection.

8. Performance Budget

• Mobile-first optimization.
• Texture sizes appropriate to device.
• Lazy loading.
• LOD where beneficial.
• Target smooth rendering at 60 FPS on supported devices.

9. Accessibility

• Provide image fallback.
• Keyboard access to essential controls.
• Respect reduced-motion preferences.
• Ensure critical product information is available outside the 3D viewer.

10. Integration

Recommended stack:
• Three.js
• React Three Fiber
• Drei
• GSAP for camera choreography
• Draco/KTX2 compression where appropriate.

11. Quality Assurance

Verify:
• Material accuracy.
• Lighting consistency.
• No texture artifacts.
• Stable interaction across supported browsers and devices.

12. Do's & Don'ts

Do:
• Keep interactions elegant.
• Optimize every model.
• Maintain realistic proportions.

Don't:
• Use excessive particle effects.
• Overcomplicate scenes.
• Sacrifice performance for unnecessary visual complexity.